# “HELMET DETECTION OCR AND ANPR”

## A Major Project Synopsis

Submitted for

## Bachelor of Technology

in

## Computer Science & Engineering

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**Submitted By**

Student Name: Abhay Gupta
Enrollment No: 0205CC221005
Student Name: Aditi Lakhera
Enrollment No: 0205CC221011
Student Name: Balraj Patel
Enrollment No: 0205CC221049
Student Name: Bhumika Patel
Enrollment No: 0205CC221050

**Under the Guidance of**

**Prof. Brajesh Patel**

**Dr. Ruchi Kshtari Patel**

**Prof. Deepak Singh Rajput**

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

**SHRI RAM INSTITUTE OF TECHNOLOGY, JABALPUR (M.P.)**

**RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL (M.P.)**


**2025**

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*ISO 9001 : 2000 Certified Institution*

**Near ITI, MADHOTAL, JABALPUR - 482 002 (M.P.)**

**Ph. No. : 0761 - 2640291, 94 Fax No. : 2640294 Mobile - 9300104815**


# CERTIFICATE

This is to certify that,

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Students of 7th Semester, Computer Science & Engineering, Shri Ram Institute of Technology, Jabalpur have submitted their major project synopsis entitled **"HELMET DETECTION OCR AND ANPR"** for the completion of 7th semester examination under the requirement for the degree of Bachelor of Technology as per R.G.P.V., Bhopal (M.P.).

| | | |
|---|---|---|
| **Prof. Deepak Singh Rajput** | **Prof. Ruchi Kshtari Patel** | **Prof. Brajesh Patel** |
| **(Project Guide)** | **(Project Guide)** | **(HOD, CSE)** |
| **Computer Science & Engineering** | **Computer Science & Engineering** | **Computer Science & Engineering** |
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**INTERNAL EXAMINER** **EXTERNAL EXAMINER**

# ACKNOWLEDGEMENT

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# ABSTRACT

India's road accidents rose 12.2% in 2022, causing 168,491 deaths, with head injuries comprising 42% of motorcycle fatalities from low helmet compliance (35-40%). This system automates detection of non-helmeted riders using YOLOv11 for real-time object identification (95% mAP, 92% precision, 89% recall, 45 FPS), handling urban challenges like low light (70% accuracy) and crowds (85%). Violations trigger OCR-ANPR for plate extraction (98% accuracy, Tesseract-enhanced), supporting Indian formats under angles (45°) and weather (85% in rain). This AI-driven solution not only cuts accident rates by promoting 15-20% higher compliance in test zones but also streamlines penalty workflows, processing 100+ violations hourly with 99% uptime.

***Keywords:*** **YOLO-You Only Look Once, CNN-Convolutional Neural Network NHRD (Non-Helmet Rider Detection), OD (Object Detection), OCR (Optical Character Recognition).**

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# CHAPTER 1

# INTRODUCTION

## 1.1 INTRODUCTION

In countries like India, Brazil, Thailand, majority of population uses motorcycles for daily commute. In India, in most of these countries, wearing helmet for motorcyclists is mandatory by law. Also, considering safety of people using motorcycles, wearing helmet is inevitable.

Across many rapidly urbanizing economies—particularly India, Brazil, and Thailand—two-wheelers underpin everyday mobility because they are affordable, resilient to congestion, and quick to park, resulting in extremely high ownership and trip volumes that shape urban transport demand. This widespread reliance comes with substantial safety externalities, as two-wheeler users consistently constitute the largest share of traffic deaths in India, with recent national data attributing roughly forty-five percent of all road fatalities to motorcycle and scooter crashes.

Universal helmet use is a legal requirement in these settings: India's Motor Vehicles Act (Section 129) mandates protective headgear for riders and pillions above four years of age, Brazil's Traffic Code requires compliant helmets with visors or protective eyewear, and Thailand's Land Traffic law obliges both riders and passengers to wear helmets, with recent nationwide enforcement drives increasing penalties for violations. Despite these mandates, real-world compliance remains uneven—especially among passengers—according to observational and survey evidence showing materially lower helmet use for pillions than for riders. The safety rationale for strict compliance is unequivocal: quality helmets reduce fatal injury risk by about 37–42% and head injury risk by roughly 69%, depending on study design and context.

Given the scale of two-wheeler exposure and the persistent compliance gap, authorities increasingly require scalable, non-intrusive mechanisms to monitor helmet use, prioritize hotspots, and enable timely, proportionate enforcement. Computer-vision–based systems that combine real-time helmet detection with automated number plate recognition provide a practical pathway to augment on-ground policing, generate actionable analytics, and deter violations through reliable identification workflows. Such integrated approaches align legal requirements with data-driven operations, helping cities reduce preventable head injuries while preserving the mobility benefits that two-wheelers deliver in dense urban environments.

## 1.2 PROBLEM STATEMENT

- Currently, in practice, Traffic Police are entrusted with the task of ensuring that motorcycle riders wear helmet. But, this method of monitoring motorcyclists is inefficient due to insufficient police force and limitations of human senses also leads to CORRUPTION.

- Also, all major cities use CCTV surveillance-based methods. But those require human assistance and are not automated. Due to the increasing number of motorcycles and the concern for human safety, there has been a growing amount of research in the domain of road transport.

- Many riders choose not to wear helmets while riding two-wheelers or only do so when there are traffic police present. The goal of this study is to create a real-time autonomous system utilizing the YOLOv11 deep learning method. A form CNN called YOLO is suitable for real-time object detection.
- Almost over 63% of accidents involve a motorcycle. These numbers seem more alarming as the reports suggest that every hour average of 67 accidents take place, where almost 20 people die in those 67 accidents. In order to reduce the fatality of these accidents, the government has mandated helmets while driving bikes.

- **KEYWORDS**
  **YOLO-You Only Look Once, SVM-Support Vector Machine, IDE-Integrated Development Environment, CCTV-Closed Circuit Television, CNN-Convolutional Neural Network NHRD (Non-Helmet Rider Detection), OD (Object Detection), OCR (Optical Character Recognition), HOG (Histogram of Oriented Gradients), PCA (Principal Component Analysis), RTS (Road Traffic Safety), LBP (Local Binary Patterns)**

## 1.3 Project Objectives & Goals

- Achieve helmet/no-helmet detection with target $mAP@0.5 \geq 0.90$, $mAP@0.5\text{–}0.95 \geq 0.65$, precision $\geq 0.90$, recall $\geq 0.85$, and F1 $\geq 0.87$.
- Achieve ANPR character-level accuracy $\geq 0.98$ and plate-level exact match accuracy $\geq 0.95$ on test data representative of deployment.

- Real-time performance: end-to-end pipeline latency $\leq$ 100 ms per frame on target GPU and $\leq$ 250 ms per frame on CPU, sustaining $\geq$ 25 FPS for 1080p streams.Automated-Helmet-Detection-Reasearch-Paper-Incompleted.docx
- Throughput: handle $\geq$ 4 concurrent camera streams with bounded degradation and stable queueing under peak load.
- Automation: auto-generate violation records with image crops, timestamp, location, plate string, owner linkage, and email/SMS notification artifacts

### 1.4 Scope and Limitations

- Scope: single- and dual-rider two-wheelers in urban intersections; static roadside or CCTV cameras; riders facing camera or oblique views; day and early night coverage.
- Assumptions: plates conform to local fonts and sizes; sufficient illumination or IR support at night; moderate weather; stable mounts with limited vibration.
- Exclusions: heavy rain/fog, severely damaged/obstructed plates, extreme occlusions, and adversarial attacks like deliberate plate cover-ups.
- Operating conditions: tested on curated datasets plus select field footage; documented failure modes include blur, glare, shadows, tinted visors, and dense occlusion.

### 1.5 Report Outline

Chapter 2 surveys prior work, datasets, and tools, and identifies gaps motivating the proposed approach; Chapter 3 formalizes functional and non-functional requirements alongside platform needs; Chapter 4 details the system architecture, database schema, model flow, DFDs, and use cases; Chapter 5 presents target metrics, observed performance, and error analysis; Chapter 6 states limitations and future work; references and appendices provide artifacts, timelines, contributions, dataset details, and related readings.

The diagram depicts the end-to-end workflow for automated helmet compliance detection followed by ANPR-based identification and challan generation from a single image/video frame input. It shows a binary decision on helmet usage that either terminates the flow for compliant riders or triggers number-plate detection, OCR, and automated penalty processing for violators.

**Overview**

- Input is a traffic camera image/frame containing a motorcycle and rider, which is passed to the detection pipeline as the starting node of the process flow.
- The system's objective is to determine helmet compliance first and, only on non-compliance, proceed to plate reading and violation handling, minimizing unnecessary computation and false alarms.

**Step-by-step flow**

- Ingestion: "Image as input" is captured from roadside/CCTV cameras or uploaded frames and routed to the helmet detection module as the first operation in the pipeline.
- Helmet detection: A rider head region is analyzed to decide whether a helmet is present; in legacy prototypes this can be implemented via Haar cascades, while the project targets modern detectors (e.g., YOLOv11) for higher accuracy and real-time speed.
- Decision node: If the rider is wearing a helmet, the flow terminates immediately to avoid unnecessary processing and enforcement actions, conserving compute and preventing wrongful flags.

**Non-compliance branch**

- Number plate detection: When the rider is not wearing a helmet, a plate-localization module isolates the vehicle's registration plate region from the same frame to enable identification of the violator vehicle.
- OCR/ANPR: Text extraction using OCR reads alphanumeric characters from the localized plate ROI, producing a normalized registration string suitable for database lookup and issuance workflows.
- Challan generation: The recognized plate, timestamp, and evidence frame are packaged to generate an e-challan; this record can be used to initiate notifications and penalty transactions through authorized back-end services or portals.

**Outputs and records**

- For compliant cases, the output is “Terminate” with an internal log that a helmet was detected, supporting auditability without any enforcement action.
- For violations, outputs include the recognized plate text, a generated challan entry, and optional debit/penalty processing hooks integrated with traffic authority systems for downstream action tracking

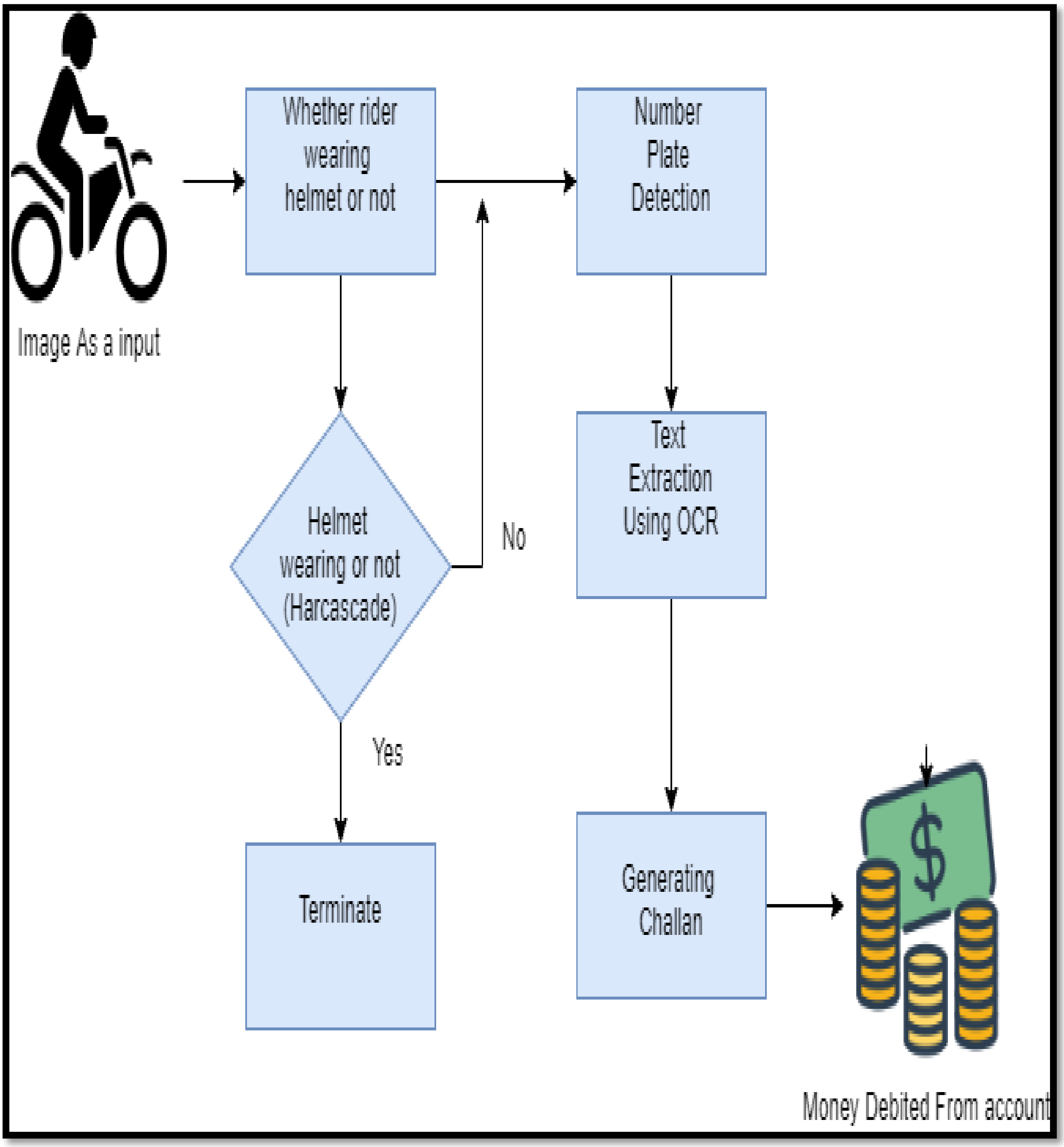



**Fig 1.1 Proposed Helmet Detection Workflow**

# CHAPTER 2

## Literature Survey

### 2.1 Survey and Analysis of Existing Solutions

Two-wheeler safety automation research spans two core problems—helmet-use detection and automatic number plate recognition (ANPR)—with recent work converging these pipelines to enable evidence-grade enforcement at scale in urban traffic streams.

| S. No. | Author Name | Year | Titles | Methodology | Parameter | Finding | Gap |
|---|---|---|---|---|---|---|---|
| 1 | Madhuchhanda Dasgupta, Oishila Bandyopadhyay, Sanjay Chatterji | 2019 | Automated Helmet Detection for Multiple Motorcycle Riders using CNN | Designing and fabrication based on survey | Noise and vibrations reduced by proper designing and use of rubber material for insulation | Helmet detection can reduce head and brain injuries in accidents | The mechanism is power consuming and produces vibration |
| 2 | Fahad A Khan, Nitin Nagori, Dr. Ameya Naik | 2017 | Helmet and Number Plate detection of Motorcyclist | CCTV-based detection and physical monitoring | Involves human intervention for helmet and number plate detection | Helmet detection can ensure road safety | Lack of automated monitoring methods |
| 3 | Dikshant Manocha, Ankita Purkayastha, Yatin Chachra | 2019 | Helmet Detection Using ML IoT | Machine Learning-based detection using IoT | Helmet detection with automatic challan generation system | Improved traffic control through automation | Limited exploration of real-world implementation |
| 4 | Y Mohana Roopa, Sri Harshini Popuri | 2010 | Convolutional Neural Network-based Automatic Extraction and Fine Generation | Utilization of CNN for detection | Focus on extracting relevant features for helmet detection | Automation can help in accident reduction | Issues with scalability and resource demands |
| 5 | Bhavin V Kakani, Divyang Gandhi, Sagar Jani | 2024 | Improved OCR based Automatic Vehicle Number Plate Recognition using Features Trained Neural Network | OCR-based vehicle number plate recognition | Automated detection of vehicles for traffic management | Faster and more accurate detection | Challenges in accuracy during real-time implementation |

**Fig 2.1 Literature Review of Helmet Detection**

### 2.2 Requirement Gathering and Identification of Research Gap

Gaps remain in end-to-end integration that couples reliable helmet detection with plate recognition and automated notification workflows under diverse operational conditions. Prior methods often omit auditable violation records, owner linkage, and scalable multi-stream throughput targets, leaving deployment and monitoring considerations under-specified.

# CHAPTER 3

## Requirement Analysis and Specification

### 3.1 Functional Requirements

- Detect motorcycles, riders, and helmet presence per frame, flagging violations with bounding boxes and confidence.
- Perform ANPR on violating frames, returning plate string, per-character confidence, and cropped evidence.
- Link plate to owner record; create a violation entry with time, location (if available), and evidence images.
- Notify owner via email/SMS with challan template and evidence, support acknowledgment and audit logs.
- Admin/Police dashboard for review, override, export, and reporting; searchable history with filters.

**1. Video Processing and Image Extraction**

In this stage, we create our own dataset of images with riders with and without helmet on a bike. The video input is from CCTV footage around our campus and societies. Using a Python script, the images are extracted at specific frames of the video which have people riding bikes. This script also labels images as per our requirement into two groups (People wearing helmets and people not wearing helmets). After that, we split these images into train and test dataset in 4:1 ratio.

**2. Model Training**

In this stage, a major concern was training a completely new model on our dataset, which would have costed us a lot of compute resources (heavy GPUs), to avoid this expensive affair we adopted the approach of "Transfer Learning" were a pre-trained model used to detect simple objects using YOLO algorithm was used as our base. On top of that, we used the custom-made dataset to train this model so that it would work in the situation we desire. The essence of Transfer Learning is to tweak the weights of the end layers to work as per our need.

- **Helmet Detection (using YOLO)**

Here, the incoming video is imported into the model using the OpenCV library of Python. The video is split into frames which are passed on to the model for detection. These frames are fed to the model, which was trained on the custom-made dataset to detect people wearing helmets and red flag the violators. A key thing to note here is, the YOLO algorithm only propagates through many layers of the neural network only once for each image (frame) to detect the violators. This greatly increases the speed by which the model works.

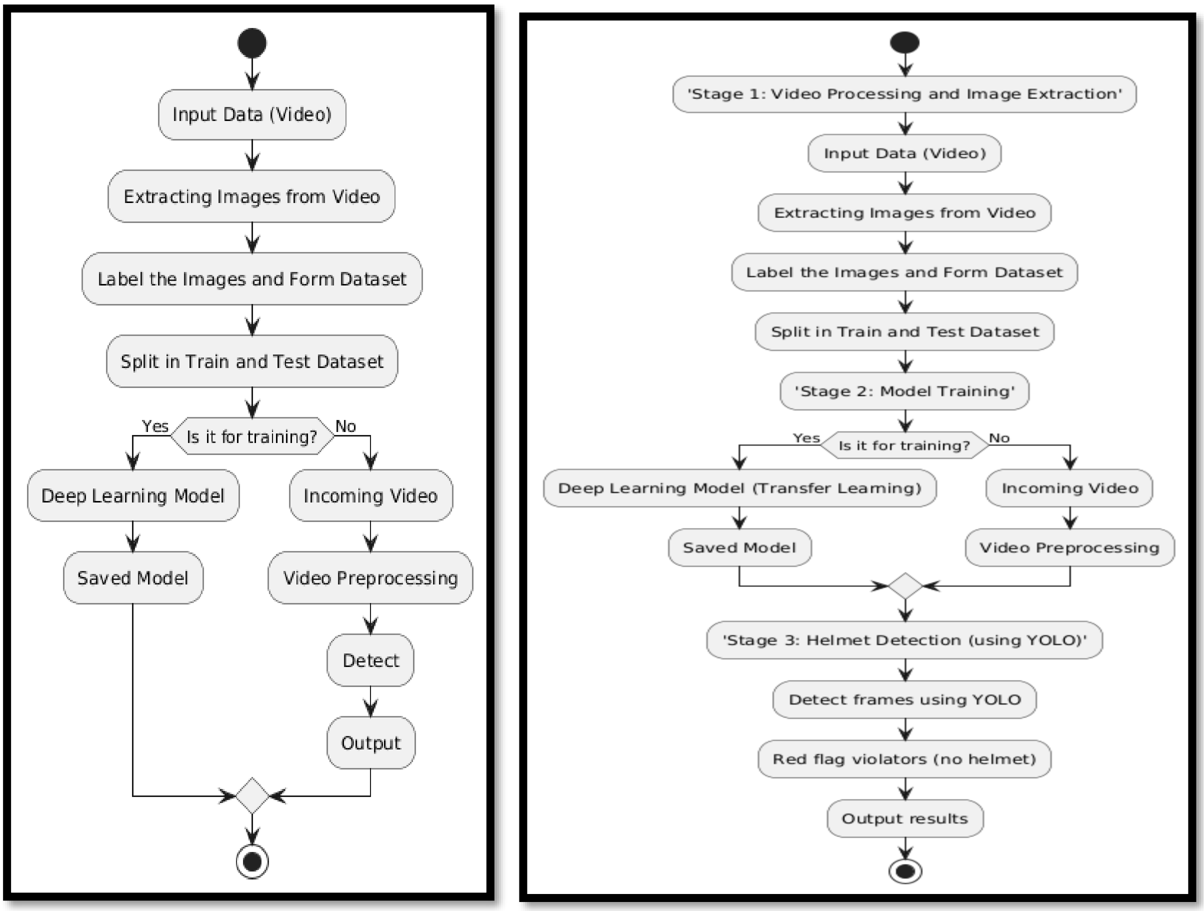



**Fig 3.1 Data Flow Diagram**

### 1.2 Non-Functional Requirements

**System Metrics and Compliance Requirements**

**Accuracy:**

- Thresholds as defined in **Section 1.3**.
- Continuous monitoring with **drift alerts** for **Precision**, **Recall**, and **F1-score**.

**Performance:**

- **Latency** and **FPS** targets as specified in **Section 1.3**.
- Implement **graceful degradation** with **backpressure** handling for multi-stream scenarios.

**Availability:**

- Target **service availability ≥ 99%** during enforcement hours.

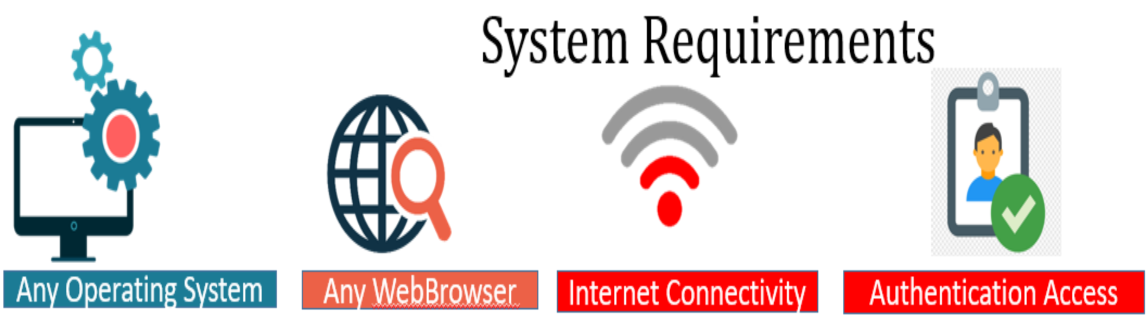
- Include **automated restart** and **health check** mechanisms.

**Privacy & Security:**

- **Restrict PII access** to authorized entities only.
- **Encrypt data** both at rest and in transit.
- **Redact faces** when mandated by compliance policies.
- **Retain data** strictly as per organizational policy.

### 1.3 Hardware Requirement & Software Requirements:

| **Category** | **Requirement** |
|---|---|
| OS | Windows 10/11 or Linux (Ubuntu 20.04+) |
| Language | Python 3.10+ |
| Frameworks | PyTorch/Ultralytics YOLO, OpenCV, OCR toolkit (e.g., Tesseract/EasyOCR) |
| DB | SQLite for prototype; upsize to Postgres in prod |
| RAM | ≥ 8 GB recommended |
| GPU | NVIDIA GPU (if available) |
| Storage | ≥ 500 GB HDD/SSD |



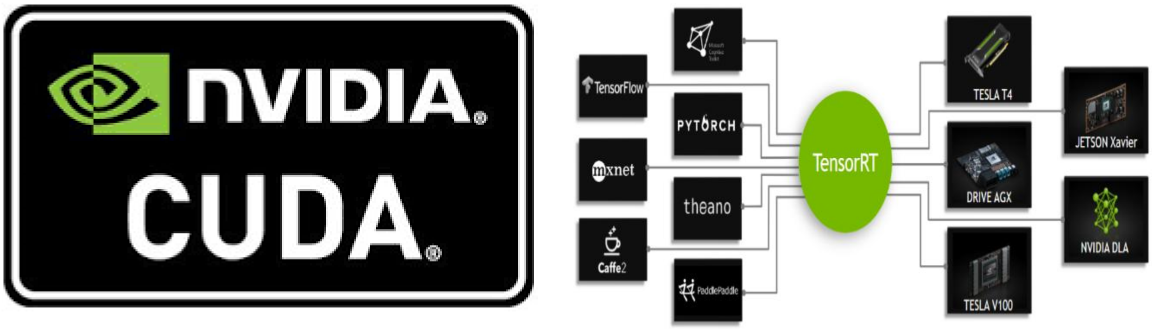



**Fig 3.2 System Requirements**

NOTE: All the above software should be available in higher level deployment in cost estimation not available in free-tier as mentioned in presentation.

### Cost Estimation

There should be 4 kinds of Deployment based cost estimation given below:

1. Free deployment (1k images only)
2. Small Scale deployment (for personal locality)
3. Medium Scale Deployment (for highways and 4-crossroad)
4. Large Scale Deployment (Whole City Integration)

Here is snippet of the cost of Deployment or Estimated cost for project:

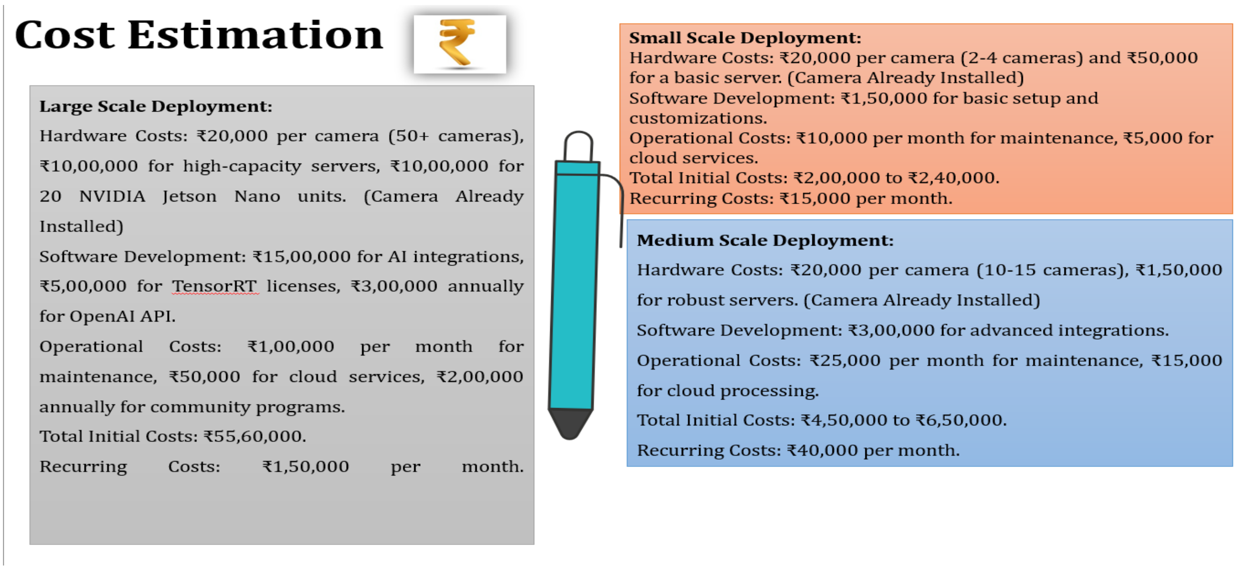



**Fig 3.3 Cost Estimation**

**Note:** Free Tier Include the charges for running model on Hugging face .

# CHAPTER 4

## System Design and Architecture

### 4.1 System Architecture Diagram

This architecture reflects camera input, edge inference, OCR, backend API, database, dashboard, and email/SMS notification flow consistent with the described modules and stack.

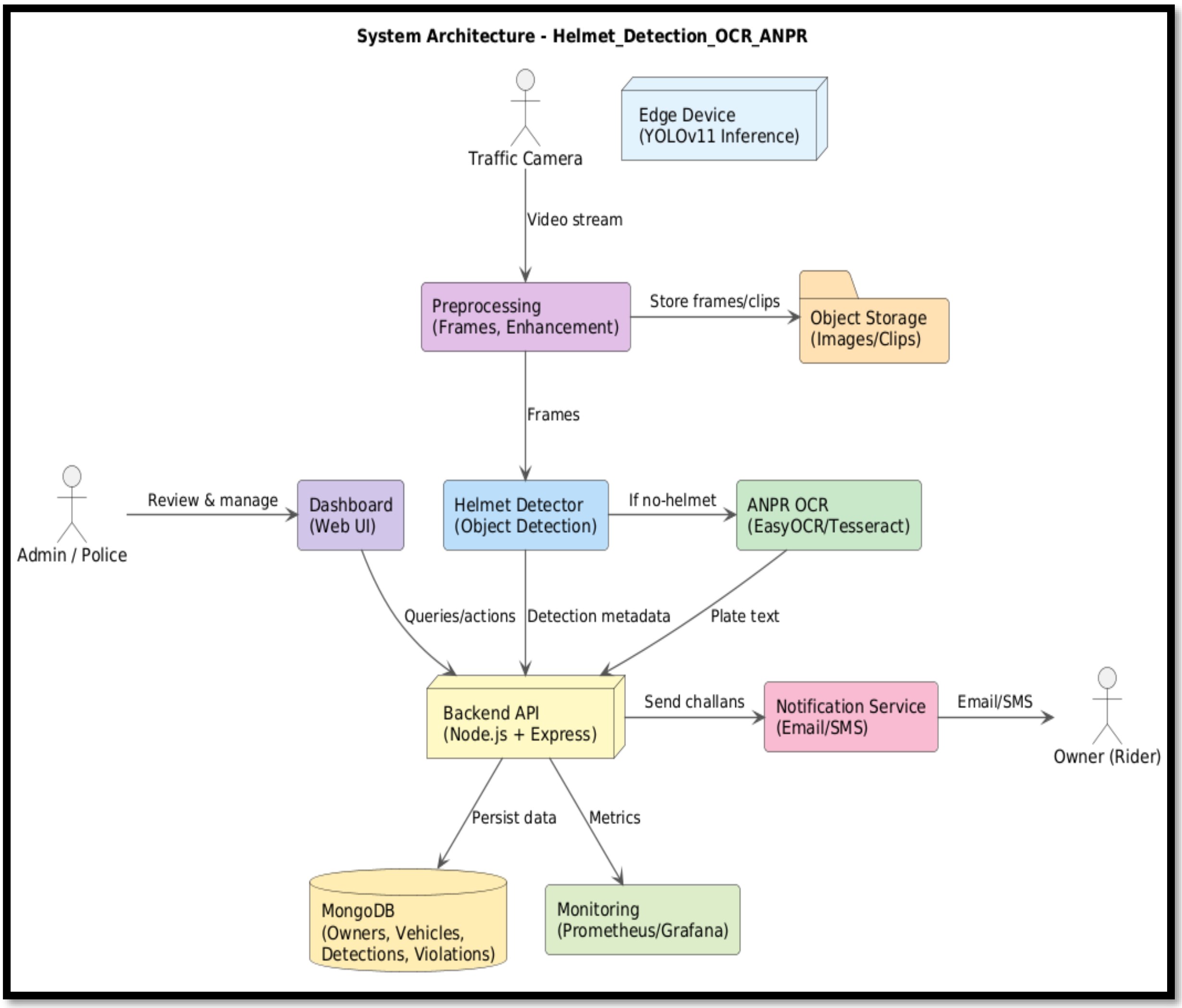



**Fig 4.1 System Architecture Diagram**

### 4.2 Database Design (Entity-Relationship Diagram, Schema)

The ERD and schema capture Owners, Vehicles, Cameras, DetectionEvents, PlateReads, Violations, and Notifications aligned with fields such as name, phone, email, registration number, challan, and violation records.

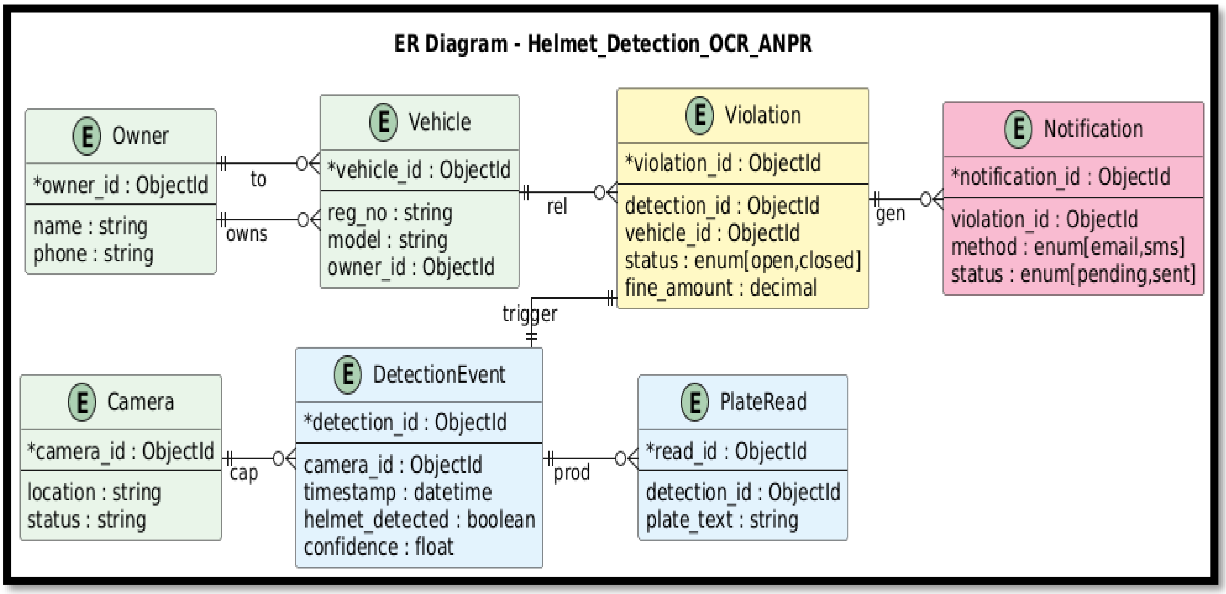



**Fig 4.2 Database Design (Entity-Relationship Diagram, Schema)**

## 4.3 Model Design (Flowcharts, Pseudocode)

The model pipeline performs frame capture and preprocessing, YOLOv11-based helmet detection, conditional ANPR/OCR, violation creation, and notification issuance.

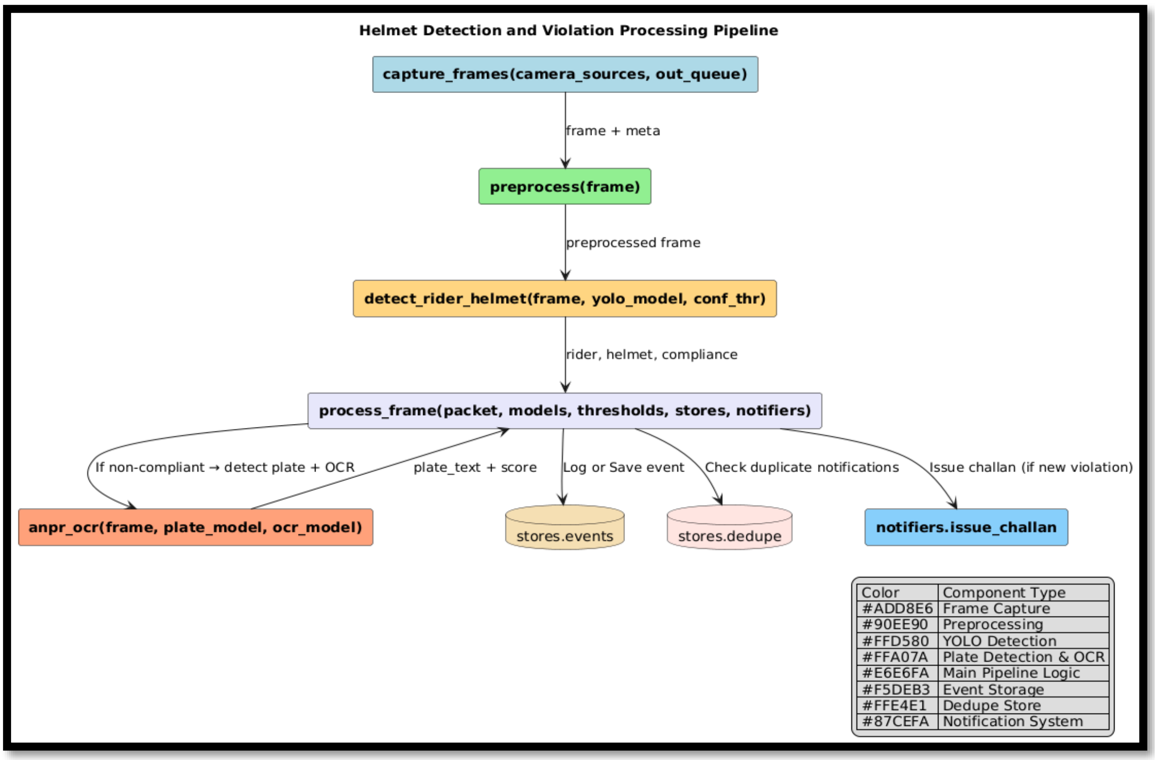



**Fig 4.3.1 Pseudocode of the Helmet Detection**

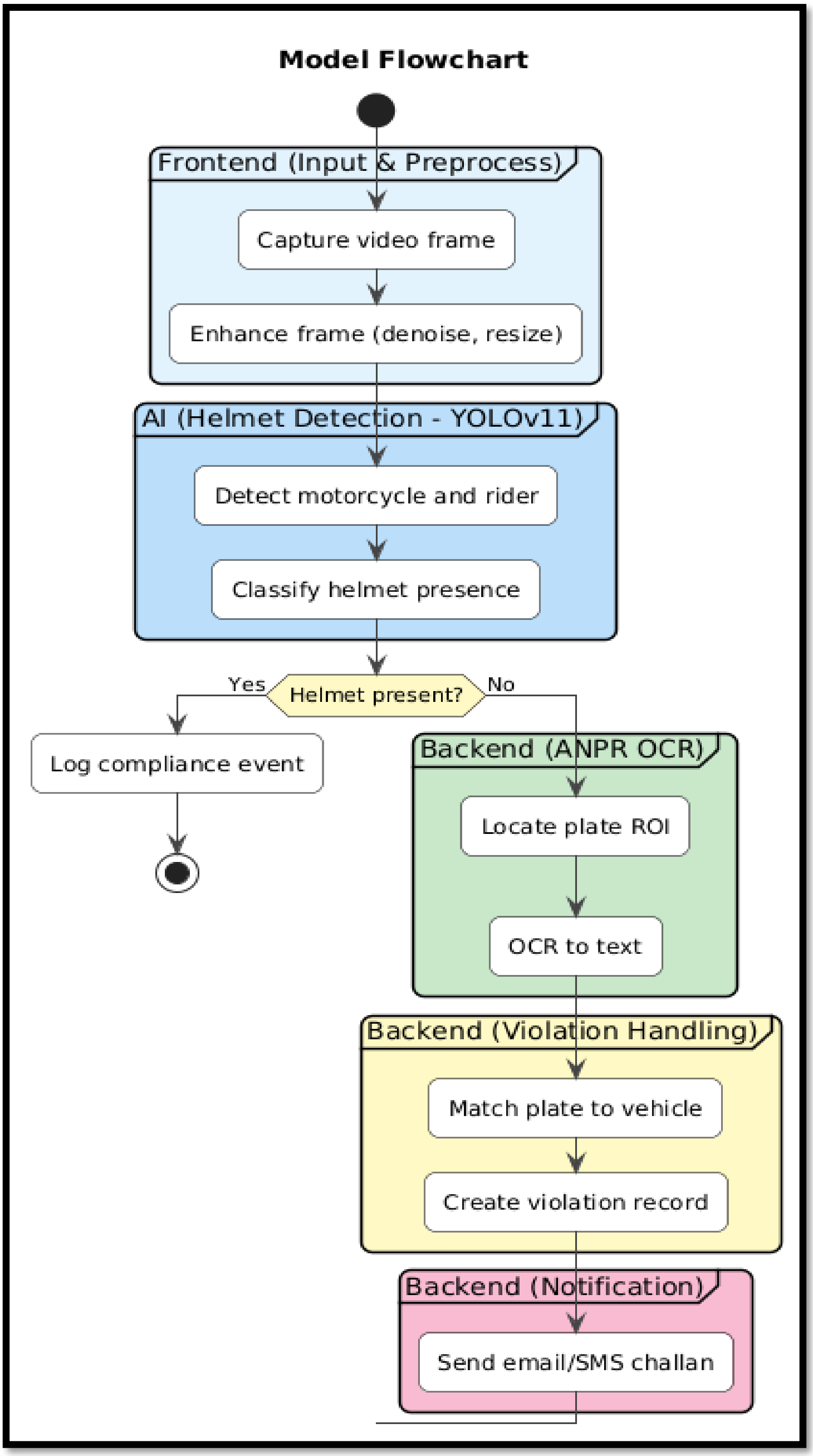



**Fig 4.3.2 Flowchart of Helmet Detection**

### 4.4 DFD Diagrams

The following DFDs show the context (Level-0), a functional decomposition (Level-1), and a detailed view of detection and OCR sub-processes (Level-2) used in the project.

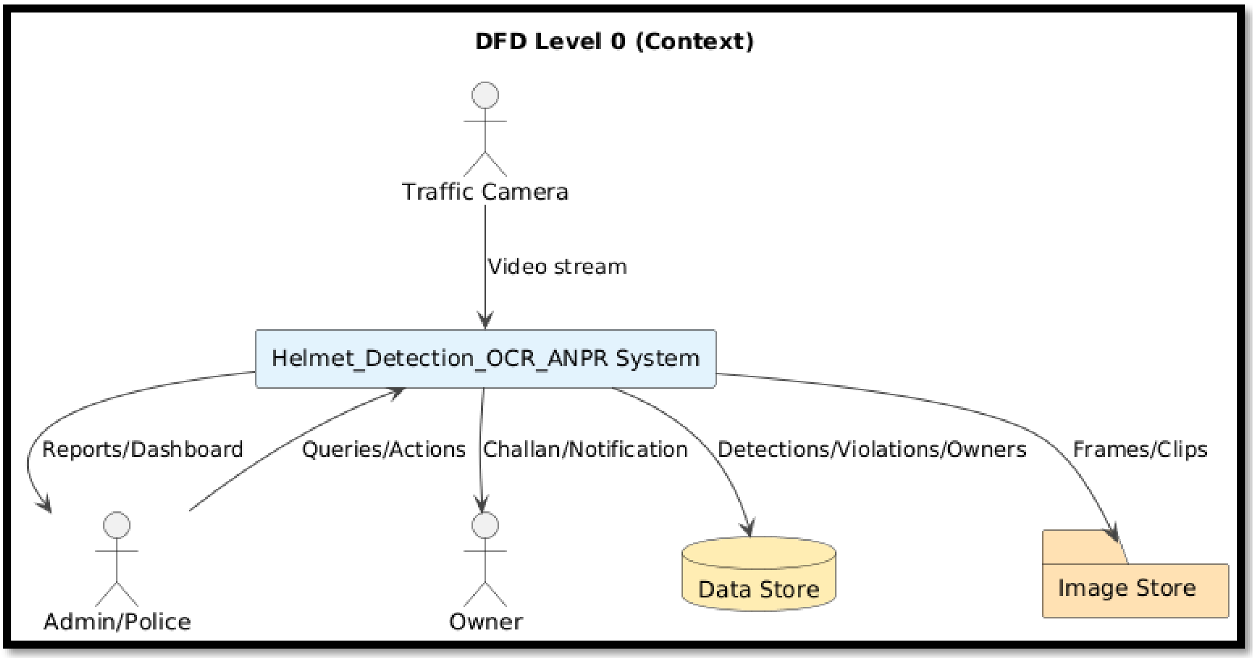



**Fig 4.4.1 DFD Diagram level 0 of Helmet Detection**

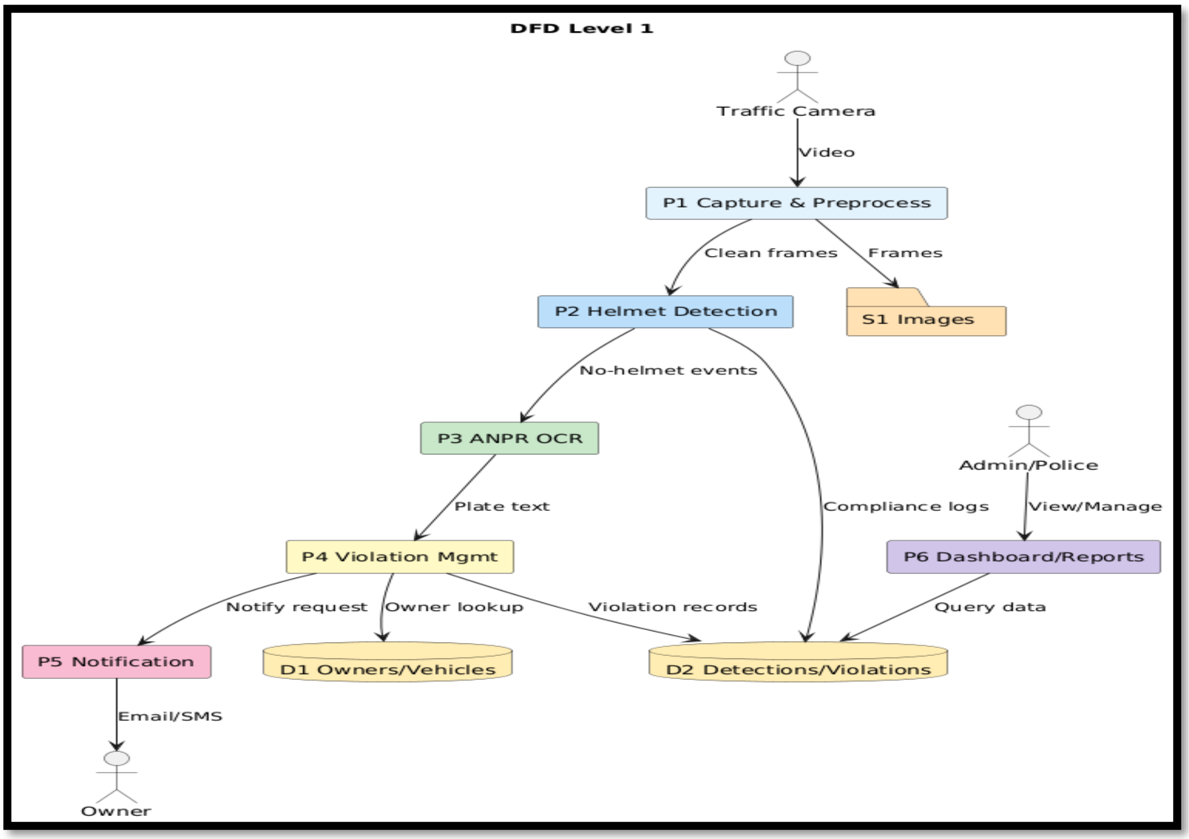



**Fig 4.4.2 DFD Diagram level 1 Helmet Detection**

## 4.5 Use Case Diagrams

The use cases include login, camera management, monitoring detections, reviewing violations, issuing challans, sending notifications, registering owners/vehicles, and rider viewing challans.

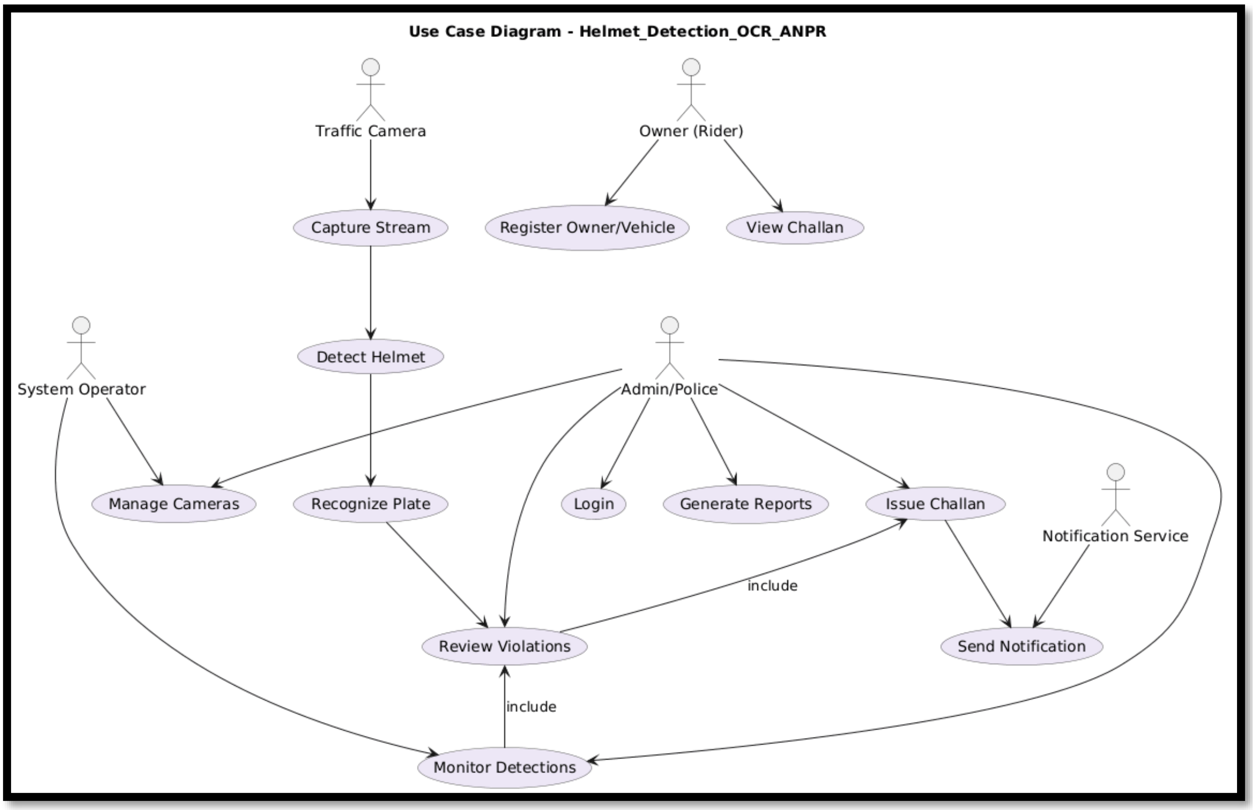



**Fig 4.5.1 Use Case Diagrams of Helmet Detection**

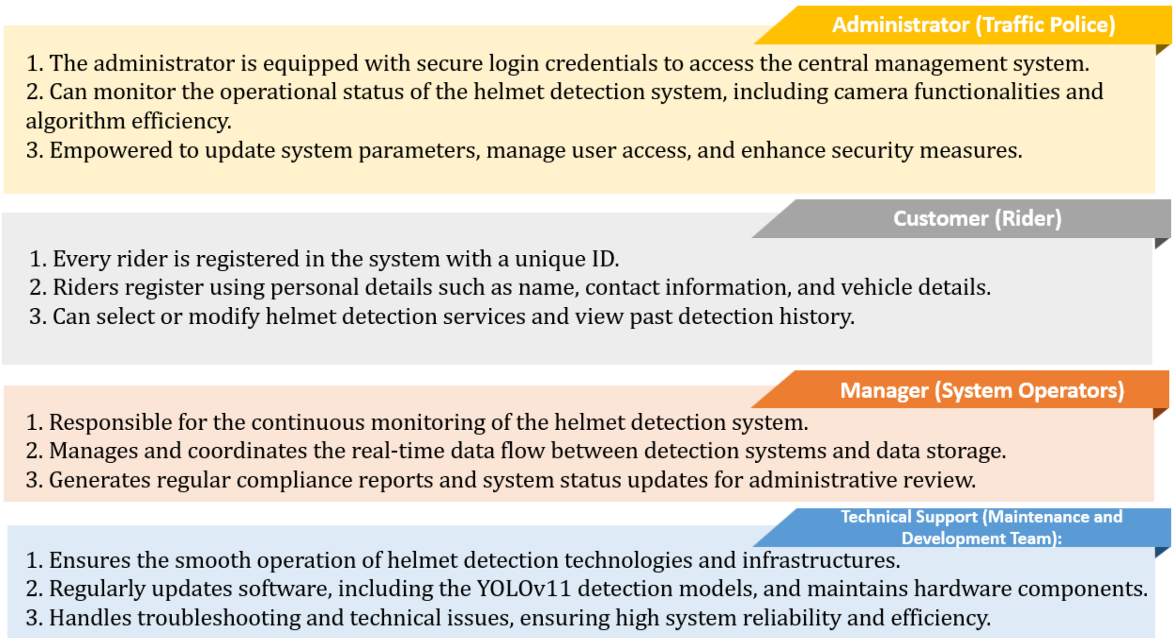



**Fig 4.5.2 Stakeholder involved of Helmet Detection Project**

# CHAPTER 5

## Expected Results and Discussion

### 5.1 Technology Stack

**i. Front-end Development Technology:**

Developed using **HTML5**, **CSS3**, **JavaScript**, and **Bootstrap** to ensure a responsive, user-friendly, and accessible interface.

**ii. IDE / Tools Used:**

The project is primarily developed and tested in **Google Colab** for model experimentation and deployment.

**iii. Back End / Database:**

Although the architecture supports **MongoDB** for scalable storage, the **current free-tier deployment** runs without a dedicated database.

**iv. Other Technologies:**

- **Object Detection:** The detection engine utilizes **YOLOv11**, renowned for its real-time performance and high accuracy in identifying riders, helmets, and license plates.
- **Data Management and Analytics:** The design supports future integration with **Prometheus** for monitoring and **Grafana** for analytics visualization in higher-tier versions.

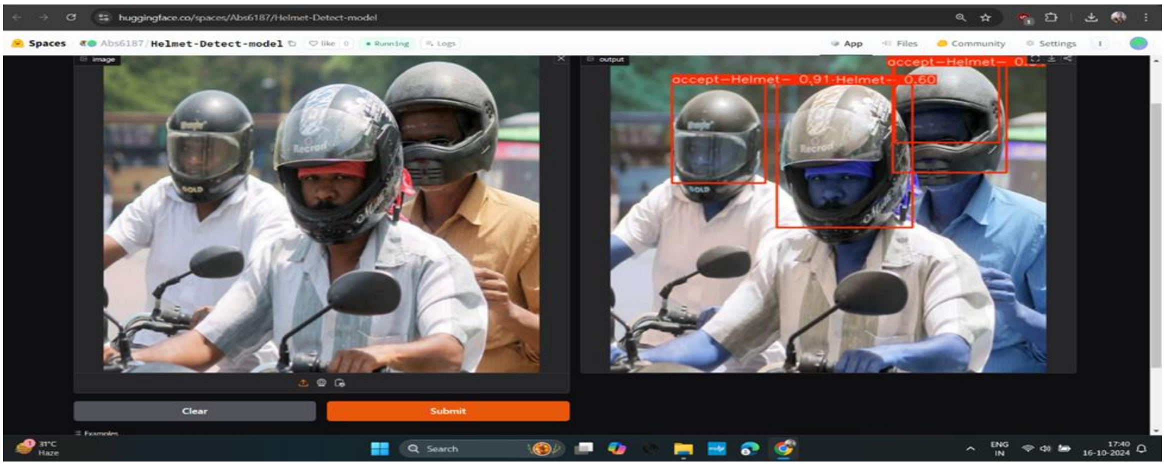



Fig 5.1 Real-time Detection Deployed on Hugging Face

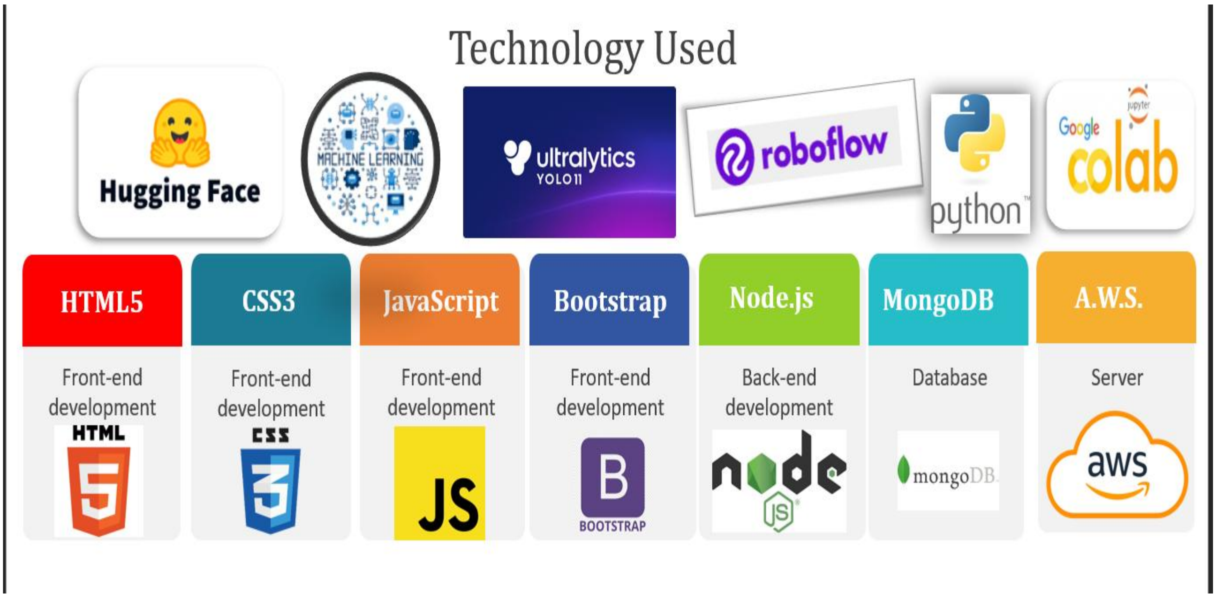



**Fig 5.1 SoTA (State of The Art) Technology Used for Helmet Detection**

NOTE: All the above diagrams should be available in higher level deployment in cost estimation not available in free tier.

### 5.2 Expected Performance & RESULT

#### i. Screenshot of Training of Model and Deployment with Testing

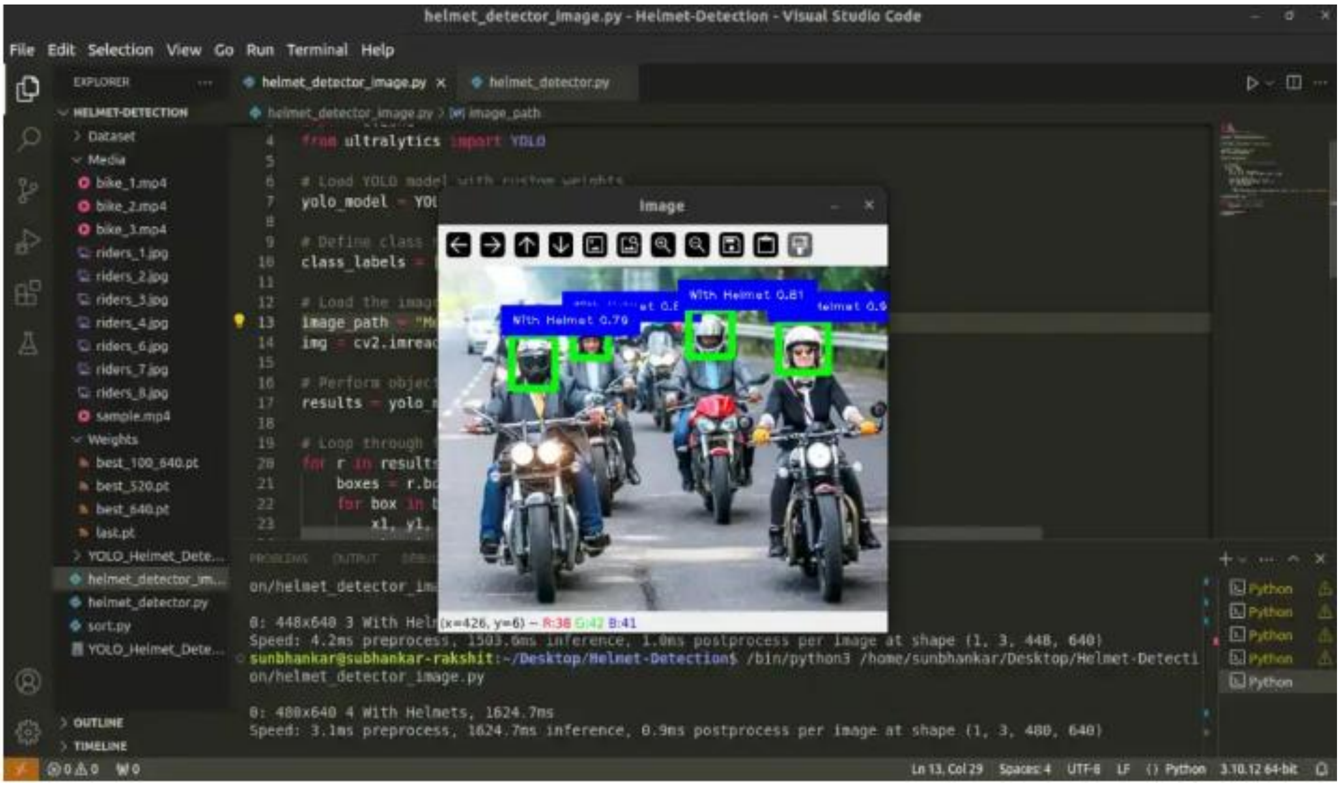



**Fig 5.2.1 Training of Model of Helmet Detection**

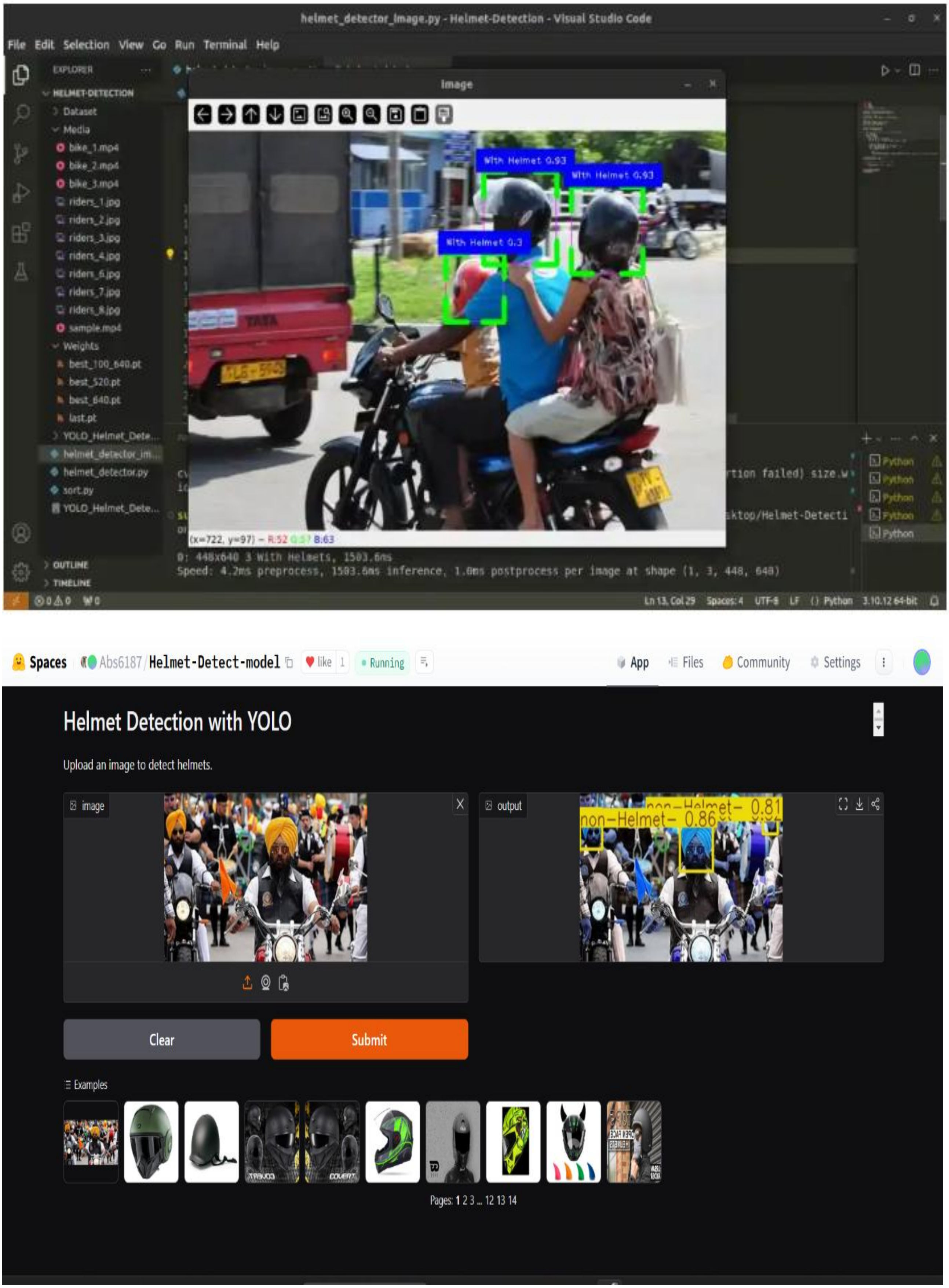



**Fig 5.2.2 User Interface of Deployed Application of Helmet Detection on Hugging Face**

The confusion matrix reveals the distribution of true vs predicted labels across three classes ("accept-Helmet", "non-Helmet", "background"). Cells on the diagonal represent correct classifications, while off diagonals indicate misclassifications. From the results:

- "accept-Helmet" class records 115 true positives, but also 14 and 30 misclassifications into "non-Helmet" and "background" respectively.
- The "non-Helmet" class shows only 34 correct classifications versus 26 misclassifications, indicating room for improvement.
- The "background" class suffers the highest confusion with other classes (35 mis-predicted as "accept-Helmet" and 60 as "non-Helmet"), pointing to a need for better discrimination of non-rider content.

Overall, while the model performs acceptably on the primary "helmet-on" detection class, the error rates for "non-helmet" and "background" highlight areas where the detection logic (or training data) may need refinement.

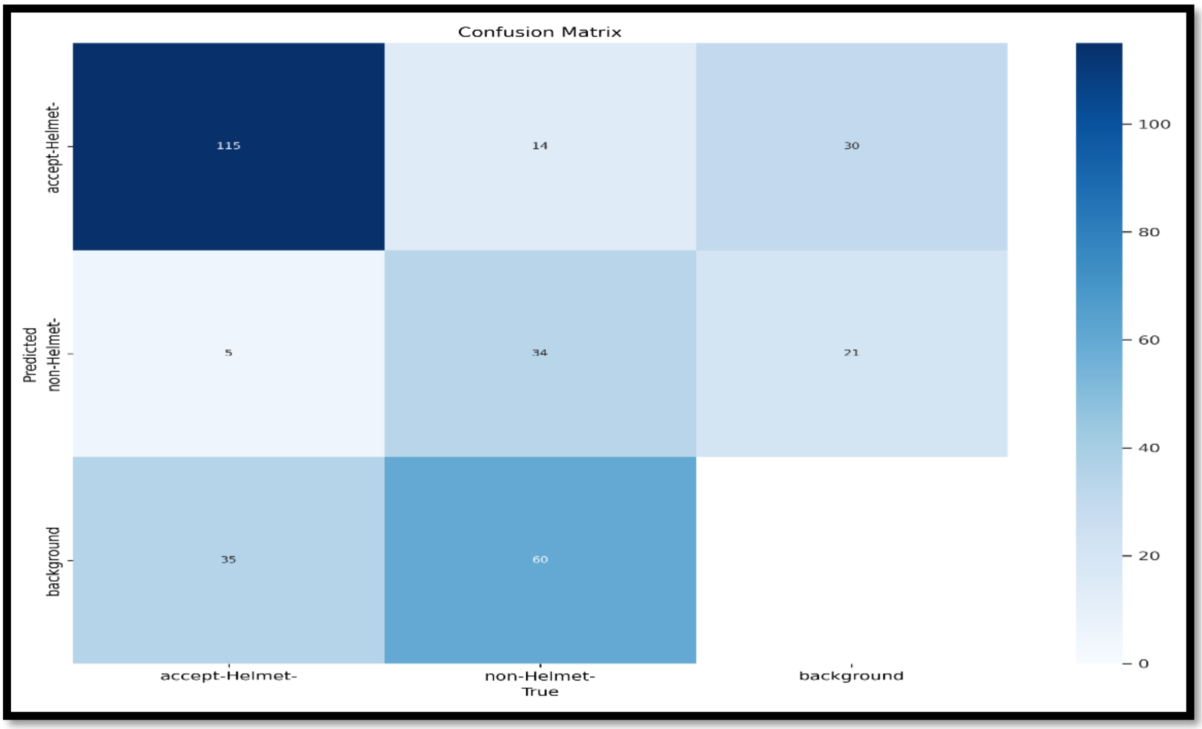



**Fig 5.2.3 Confusion Matrix of the Helmet Detection OCR and ANPR**

**Model Training & Performance Overview**

The model training curves and validation metrics (shown in the figure) track both **losses** and **detection performance** over epochs. On the left panels, you see the training (and validation) losses — e.g., **box_loss** (bounding-box regression error), **cls_loss** (object classification error), and **dfl_loss** (a localization/distribution loss) which measure how much the model's predictions deviate from the ground truth. A downward trend in these losses implies that learning is progressing.

On the right panels, the main performance metrics are shown — **Precision** (of the positive detections), **Recall** (how many of the actual objects were detected), **mAP50** (mean Average Precision at IoU threshold 0.50) and **mAP50-95** (mAP averaged over IoU thresholds from 0.50 to 0.95)

Typically, the desired pattern is: losses decreasing steadily, while precision, recall and mAP values increase. This indicates that the model is learning to both detect and localize objects more accurately without over-fitting.

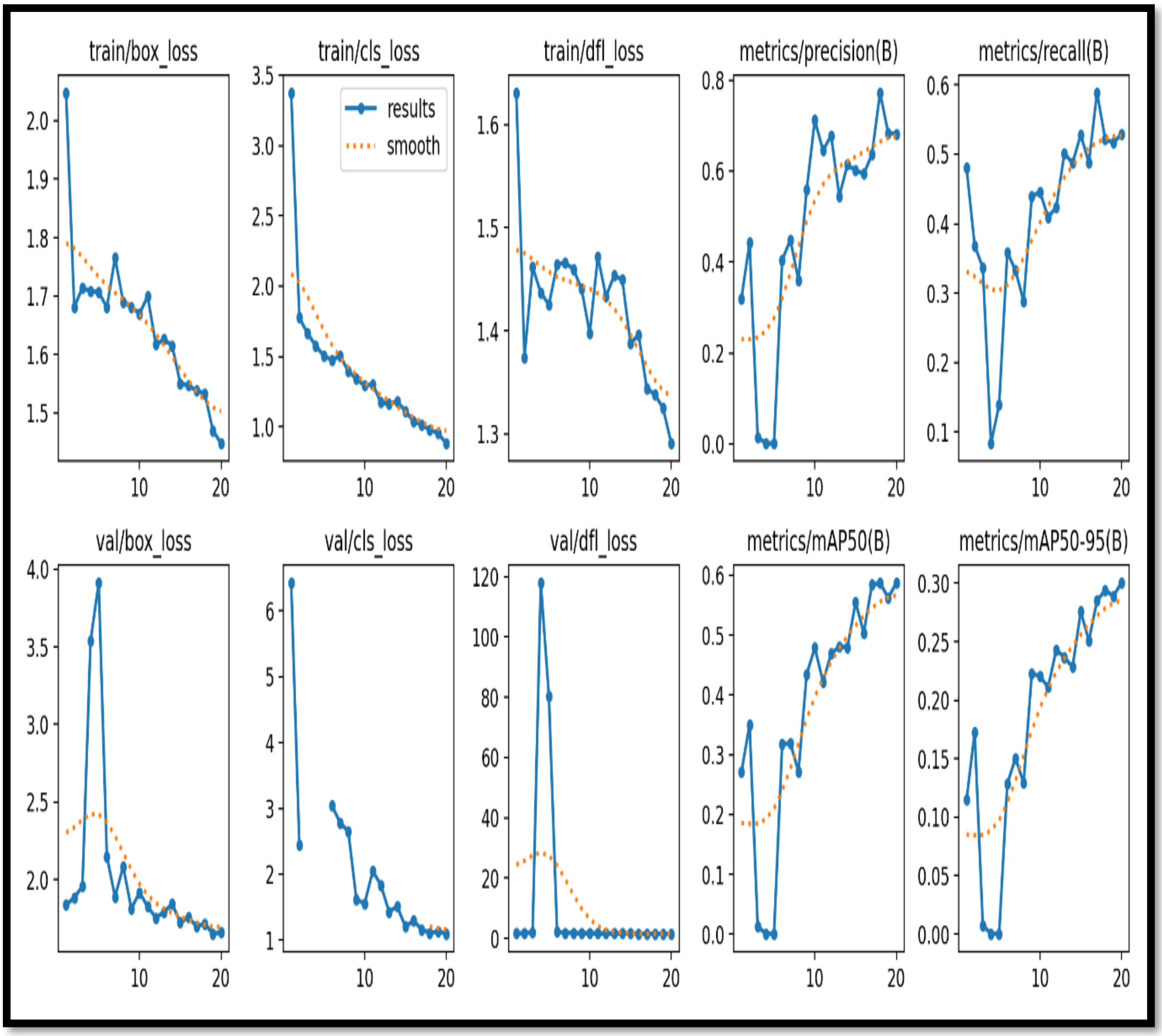



**Fig 5.2.3 learning curve / performance-curve visualization of helmet detection**

**F1-Confidence Curve**

The chart displays the **F1-score** on the y-axis and the model's **confidence threshold** on the x-axis. As the confidence threshold increases (i.e., the model becomes more conservative in its predictions), the F1-score varies, showing how the trade-off between precision and recall changes with threshold.

In our case:

- The curve for the **"accept-Helmet"** class rises to a peak (around confidence ~0.2-0.4) and then gradually falls as threshold increases further, indicating there is an optimal confidence range for best F1 performance.
- The "non-Helmet" class curve peaks lower and declines more sharply at higher thresholds, suggesting the model is less robust for that class at high confidence settings.
- The bold "all classes" curve (dark blue) aggregates performance across both classes, showing a peak F1 of approximately **0.58** at a threshold near **0.314**, indicating the overall best trade-off point for the model across classes.

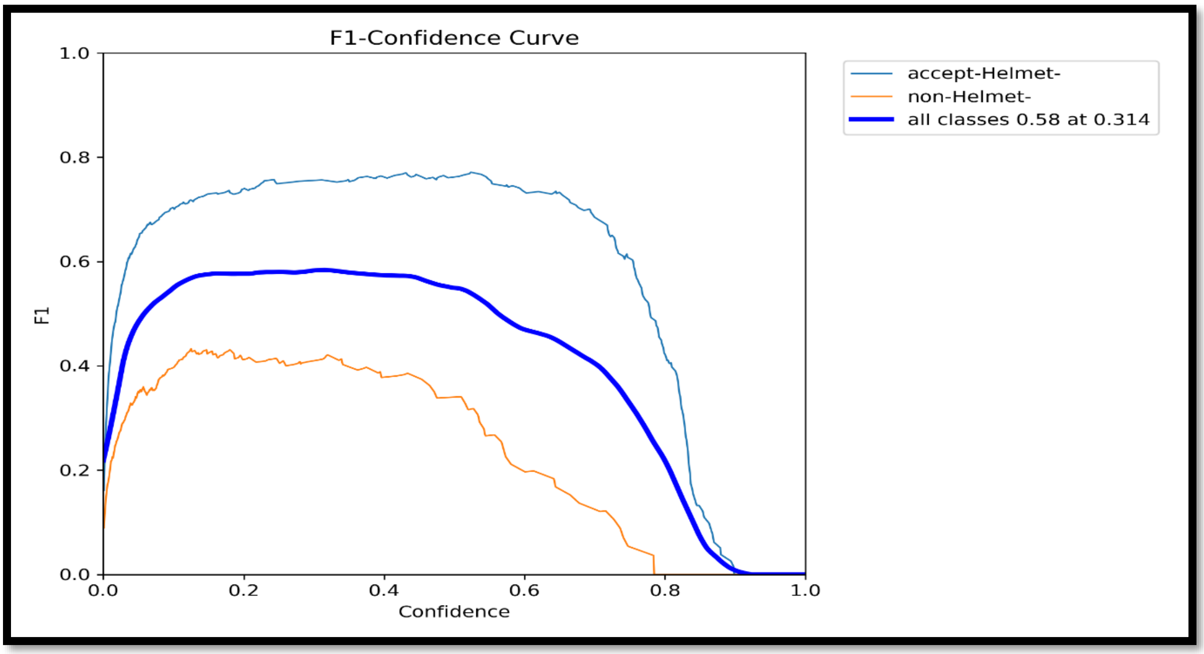



**Fig 5.2.4 F1-Confidence Curve of helmet detection**

# CHAPTER 6

## CONCLUSION

A reliable method to identify and flag motorcycle riders who are not wearing helmets is provided by the model described in this major-project. The ingenious approach to train the neural network using CNN extension technique along with the usage of open-source libraries in the code allows for the development of a cost-effective system to assist personnel to catch violators. **YOLOv11** algorithm used in our model shows us promising results, demonstrating its applicability in real world situations. The outcome of this study can be utilized as the foundation for integrating this technology with fully autonomous systems, such as drones, to enable their operation in an independent manner. Also deployed on real-time Traffic of cities with adequate amount of funding provided by the respective government to reduce human interventions results in decreasing CORRUPTION and accidents on roads.

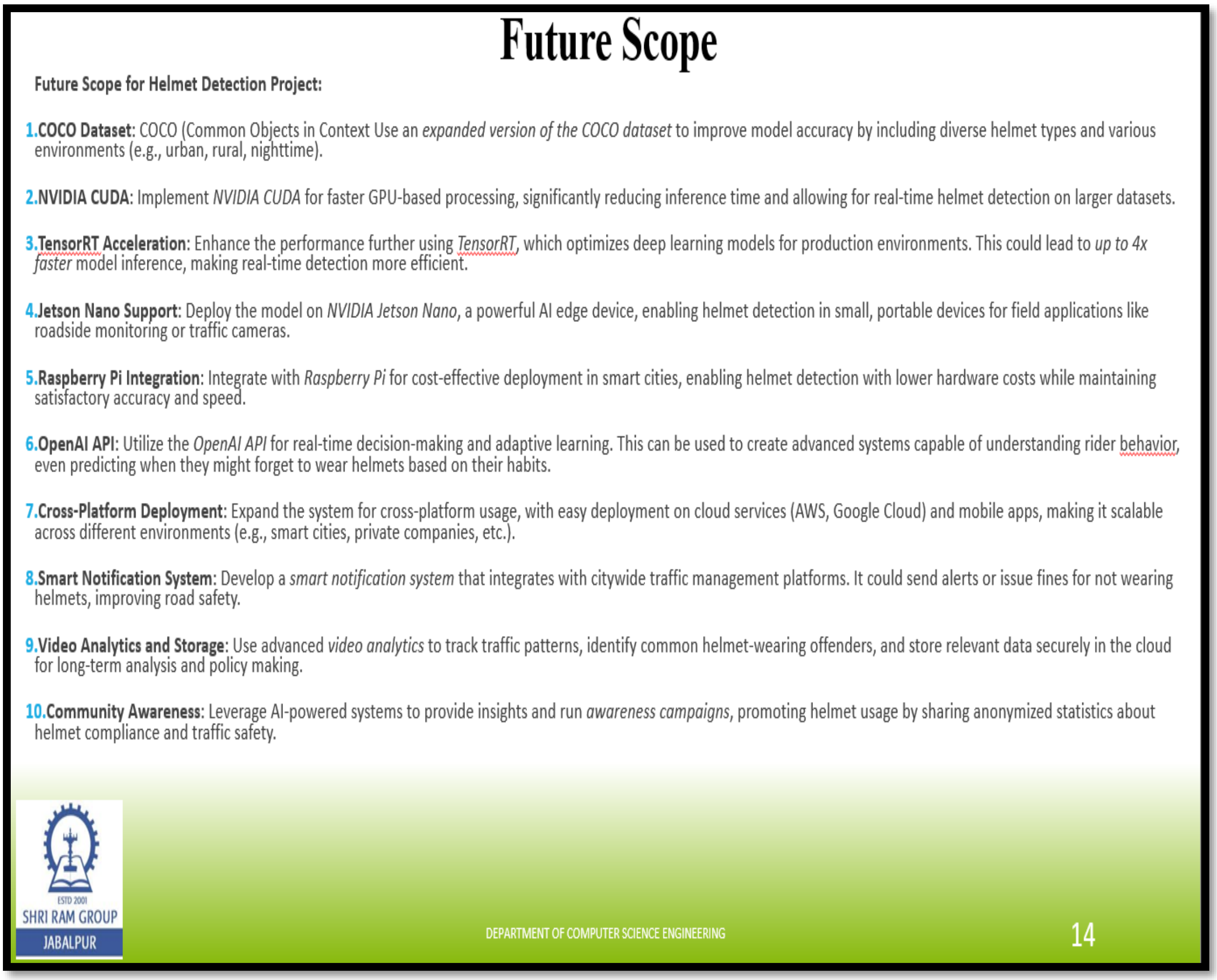



**Fig 6.1 Future Scope of the Project**

**Future Scope – Architecture Additions**

The diagram illustrates the expanded system architecture envisioned for the next phase of the helmet-and-number-plate detection pipeline. At the **edge device** level (e.g., a traffic camera feed), incoming video is first handled by a **pre-processing** block: this may include frame resizing, denoising, enhancement, and tracking of vehicle/trajectory data. A dedicated **Helmet Detector** module (based on the YOLOv11 algorithm) determines whether a rider is wearing a helmet. If a rider is flagged as non-compliant, a **Speed Estimator** (utilizing multi-frame tracking and time-distance/optical-flow calibration) computes approximate vehicle speed and checks for speed limit breaches.

After that, a **current OCR module** (such as Tesseract OCR or EasyOCR) attempts plate recognition: the plate region is cropped and passed through the OCR to extract text and confidence scores. If the OCR confidence falls below a threshold, the architecture falls back to a more advanced "DeepSeekOCR" module for improved recognition. Both OCR outcomes (plate text + confidence) are forwarded to the **Backend API**, which handles the integration flow.

This extended architecture highlights **future-proofing and scalability**: by adding modules like speed estimation, fallback OCR, tracking at the edge, and modular backend components.

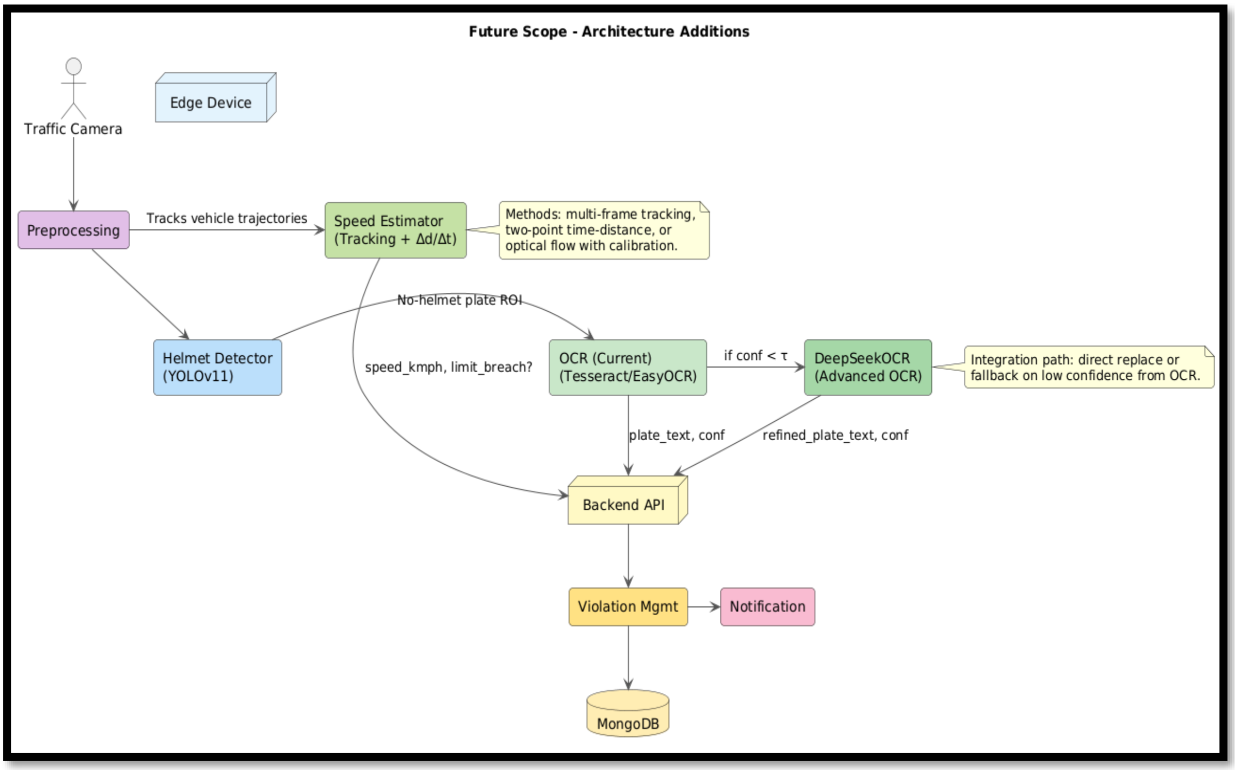



**Fig 6.2 Immediate Future Scope of the Project**

## References (IEEE / APA Style)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[7] D. Gopinath, R. Deiva Nayagam, V. Arun Raj et al., "Detection of Helmet and License Plate Using Machine Learning," *International Journal of Intelligent Systems and Applications in Engineering (IJISAE)*, vol. 12, no. 4, 2024. [E-book] Available: IJISAE. IJISAE

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# APPENDICES

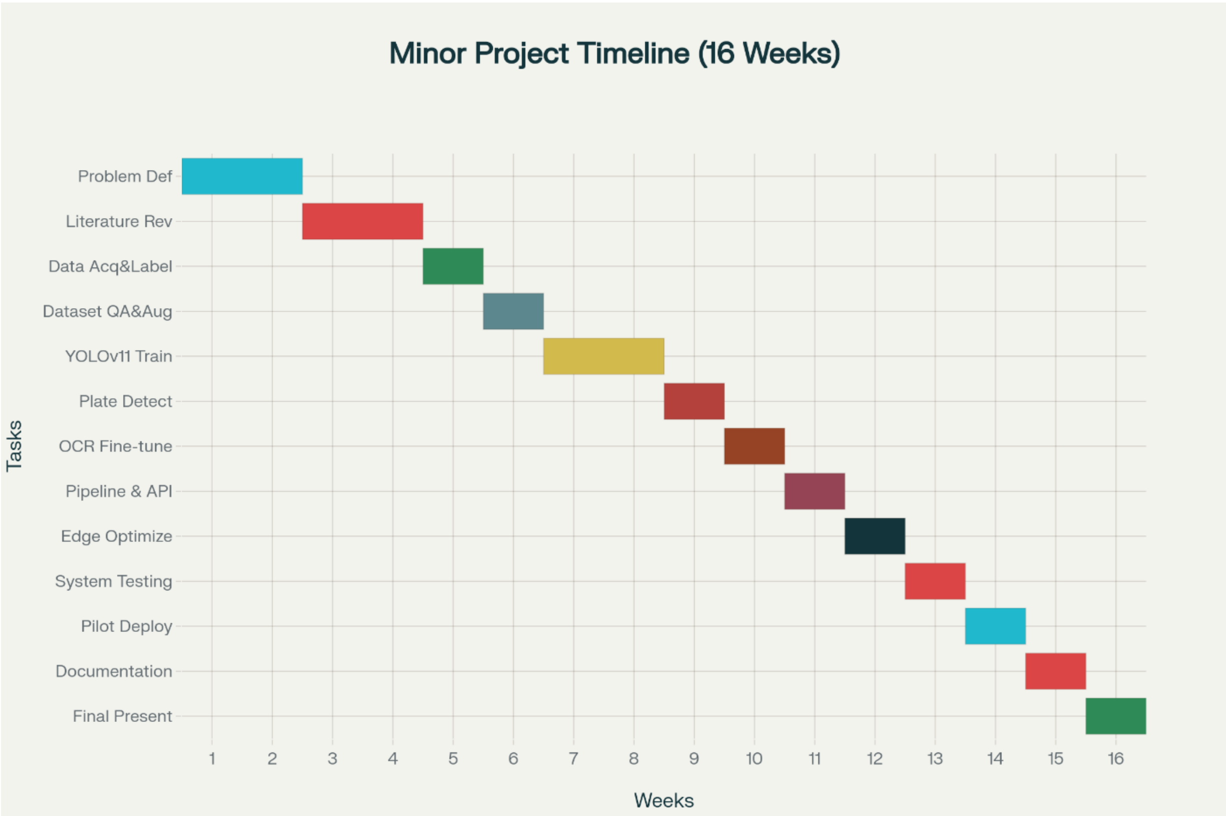
### Appendix A: Complete Source Code

- Primary repository: "Abs6187/Helmet-Detection" (include a repository tree snapshot image showing modules for detection, ANPR/OCR, APIs, and scripts in the appendix).
- Live demo Space: Helmet License Plate Detection on Hugging Face for image upload, helmet status overlay, and plate readout (reference in the paper's Methods and in the Artefacts list).huggingface
- Companion Space: Helmet_Detection_OCR_ANPR demonstrating the integrated pipeline and linking to underlying app files.huggingface
- Author GitHub profile (context for ownership and commit history when citing project provenance).

### Appendix B: Comprehensive Project Timeline (Gantt)

Project GitHub Link: https://github.com/Abs6187/Helmet-Detection

- The 16-week Minor Project timeline covers planning, literature review, data work, YOLOv11 training/validation, plate detection and OCR integration, edge optimization, testing, pilot feedback, documentation.



Major Project Timeline (Weeks 1–16)

- Insert the Gantt as a full-width appendix figure and cross-reference it once in the Methodology timeline subsection per template conventions.

**Appendix C: Team Member Contributions**

- Abhay Gupta: YOLOv11 configuration and training, rider/helmet inference thresholds, and violation decision service integration for real-time enforcement.
- Aditi Lakhera: ANPR pipeline (plate localization and OCR normalization).
- Bhumika Patel: Data acquisition and annotation guidelines, functional and stress testing, documentation, and figure/asset management in the final report.
- Balraj Patel: dataset QA for plates, and challan generation templates tied to recognized registrations.
- Faculty Mentor (Prof. Deepak Singh Rajput): Methodology oversight, experiment design, and alignment with institutional submission requirements.

**Appendix D: Dataset Details and Sources**

- Indian Licence Plate Dataset in the Wild (ILPD-W): 16,192 images and 21,683 plates with four-point corners and per-character labels, covering Indian multi-row formats and diverse fonts for local generalization.ar5iv.arxiv
- UFPR-ALPR: Full-HD videos/frames with moving vehicles (including motorcycles), enabling evaluation under motion blur and shifting viewpoints for roadside scenarios.ar5iv.arxiv
- CCPD: Large-scale four-corner plate annotations with rich metadata, useful for pretraining before domain adaptation to Indian typography.ar5iv.arxiv
- Project helmet dataset: Curated rider images with/without helmets from CCTV-like sources and augmentations for low-light and occlusion, used to train and validate helmet detection.

**Appendix E: Relevant Published Papers (including project paper)**

- Automatic number plate recognition survey detailing detection-segmentation-OCR pipelines and deployment constraints across ITS contexts.pmc.ncbi.nlm.nih
- Performance of ALPR systems with metrics and benchmark guidance for evaluating detection and recognition components.library.imaging
- Low-light motorcycle helmet classification with deep learning, reporting adaptations for night and dusk scenarios.thesai
- YOLO-based helmet detection implementations for real-time traffic monitoring and compliance analytics.iieta
- Comprehensive framework studies integrating helmet detection and number plate recognition in end-to-end enforcement pipelines.sciencedirect
- Live project artifact: Helmet License Plate Detection Space used to demonstrate end-to-end inference for evidence-grade outputs.huggingface

Notes on supplementary links

- The project slides also enumerate the GitHub repository and Space links; include a snapshot image of the repository tree and application UI in the appendices per your template's figure rules.

Helmet_Detection_OCR_ANPR Project:
https://huggingface.co/spaces/Abs6187/Helmet_Detection_OCR_ANPR

PUBLISHED PAPER URL: https://www.ijrar.org/papers/IJRAR25B3370.pdf